

SHARIF  
UNIVERSITY OF TECHNOLOGY



---

# Electromagnetic

## *Assignment 2*

---

Aria Fateh Kerdari  
403110382

May 5, 2026

## 1 Problem 1

$$A = 4R^2 \cos\theta \sin\varphi$$

$$\nabla A = 8R \cos\theta \sin\varphi \hat{r} - 4R \sin\theta \sin\varphi \hat{\theta} + 4R \cot\theta \cos\varphi \hat{\varphi}$$

$$\nabla \cdot \nabla A = 24 \cos\theta \sin\varphi - 8 \cos\theta \sin\varphi - \frac{4 \cot\theta}{\sin\theta} \sin\varphi$$

$$\nabla \cdot \vec{B} = 0 + \frac{\cot\theta \sin\varphi}{R}$$

برای مختصات کروی

$$\nabla A = \frac{\partial A}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial A}{\partial \theta} \hat{\theta} + \frac{1}{r \sin\theta} \frac{\partial A}{\partial \varphi} \hat{\varphi}$$

$$\begin{aligned} \nabla \cdot \vec{B} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 B_r) + \frac{1}{r \sin\theta} \frac{\partial}{\partial \theta} (\sin\theta B_\theta) \\ &\quad + \frac{1}{r \sin\theta} \frac{\partial B_\varphi}{\partial \varphi} \end{aligned}$$

## 2 Problem 2

$$\oint_S \vec{F} \cdot d\vec{S} = \int_V \nabla \cdot \vec{F} d\tau$$

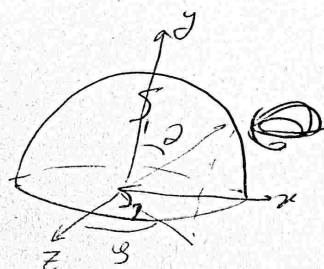
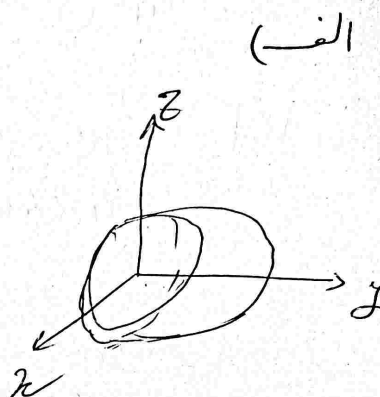
$$\nabla \cdot \vec{F} = 1$$

$$\oint_S \vec{F} \cdot d\vec{S} = \frac{2}{3} \pi R^3$$

$$n_1 = \frac{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}{R} \quad n_2 = -\hat{y}$$

$$\int_{S_1} \vec{F} \cdot d\vec{S} = \int_{S_1} \vec{F} \cdot \hat{r} dS$$

$$= \int_{S_1} \frac{4x + x^2}{R} dS, \quad x = R \sin\theta \cos\phi$$



$$\bar{I}_1 = 4R \int \frac{\sin\theta \sin\varphi \sin\theta d\theta d\varphi}{r/R} = 0$$

$$\begin{aligned} \bar{I}_2 &= R^3 \int \sin^2\theta \sin^2\varphi \sin\theta d\theta d\varphi \\ &= 4\pi R^3 \int_0^{\pi/2} \sin^3\theta d\theta = \frac{2}{3} 4\pi R^3 \end{aligned}$$

$$\int_V \vec{F} \cdot d\vec{S} = \bar{I}_1 + \bar{I}_2$$

همانطور که انتظار داریم در انتگرال با هم برابر هستند و البته  $\int_V \nabla \cdot \vec{F} dV$  بسیار راحت قابل محاسبه است.

## 3 Problem 3

$$\oint_S \vec{F} \cdot d\vec{S} = \int_V \nabla \cdot \vec{F} d\tau$$

$\hat{x}$ :

$$\vec{F} \cdot \hat{x} = x^2$$

$$\oint_{S_1} \vec{F} \cdot \hat{x} dS = - \int_0^1 x^2 dx = -1/3$$

$$\oint_{S_2} \vec{F} \cdot \hat{x} dS = 1/3$$

$\hat{y}$ :

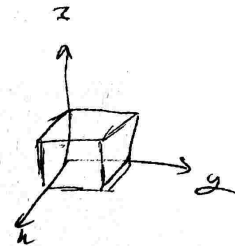
$$\oint_{S_3} (2xy + z^2) dx dz = -1/3$$

$$\oint_{S_4} (2xy + z^2) dx dz = 1 + 1/3 = 4/3$$

$\hat{z}$ :

$$\oint_{S_5} (2xz) dy = 0 \quad \Rightarrow \quad \oint_S \vec{F} \cdot d\vec{S} = 2$$

$$\oint_{S_6} 2z dy dx = 1$$



$$\oint \nabla \cdot \mathbf{F} d\tau$$
$$= \int (2x + 2z) dx dy dz = 1 + 1 = 2$$

## 4 Problem 4

$$\nabla \times \mathbf{F} = \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{1}{r} \right) \hat{z} = 0 \quad r \neq 0$$

$$\oint_C \mathbf{F} \cdot d\mathbf{l} = \oint \frac{1}{a} a d\theta = \frac{2\pi}{a}$$

به ظاهر قضیه استوکس برقرار نیست! به این خطرات که  
 شرایط قضیه برقرار نیست. در نقطه  $r=0$  (یعنی  $z=0$ )  
 $\nabla \times \mathbf{F}$  تعریف نشده است و این عامل مشکل است.

$$\nabla \times \mathbf{F} = \delta_r \hat{z} \quad \text{اگر فرض کنیم}$$

$$\iint \delta_r ds = 2\pi \int \delta_r dr = 2\pi$$

قضیه استوکس برقرار می شود.